

Capstone Project 1 - Mutual Fund Analytics

Industrial-Grade ETL Pipeline, Exploratory Data Analysis,
Risk-Adjusted Performance Analytics & Dashboard Design

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Deliverables: Final_Report.pdf | Presentation.pptx | GitHub Repository (v1.0)

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1. Executive Summary

This report presents the complete deliverable for Capstone Project 1 - Mutual Fund Analytics at Bluestock Fintech Pvt. Ltd.. The project implements an automated analytics pipeline processing 29,125 NAV records (Dec 2012 to Jun 2026) across 6 large-cap equity schemes, combined with 32,778 investor transaction records for behavioural analysis.

Key Findings

- Highest CAGR: Axis Bluechip Direct at 51.1% over the full observation period.
- Best risk-adjusted return: HDFC Top 100 Direct (Sharpe = 0.73).
- SIP accounts for 60.2% of all transactions, confirming retail preference for systematic investing.
- Industry SIP inflows reached an all-time high of Rs 31,002 Cr in December 2025.
- SBI Mutual Fund leads AMC AUM at Rs 12.5 Lakh Crore (2025), per industry dataset.

Objectives Completed (Days 1-7)

- Project structure, Git repo, requirements.txt, and 10 CSV datasets ingested
- Live NAV fetch from mfapi.in for 6 anchor schemes
- SQLite star schema with 10 analytical SQL queries
- 15+ EDA charts exported to reports/charts/
- Performance metrics: CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Scorecard
- Advanced analytics: VaR/CVaR, rolling Sharpe, cohort & SIP continuity analysis
- Power BI dashboard design (4 pages) with data connection documented
- Final report (this document) and 12-slide presentation

2. Data Sources & Methodology

2.1 Primary Sources

NAV data was sourced from the public MFAPI endpoint (https://api.mfapi.in/mf/{scheme_code}) and consolidated with AMFI-style reference datasets. Investor transactions are synthetically generated for EDA demonstration with full demographic attributes.

2.2 Fund Universe (Six Anchor Schemes)

- HDFC Top 100 Direct (AMFI 125497) - HDFC Mutual Fund, Large Cap, Very High
- SBI Bluechip (AMFI 119551) - SBI Mutual Fund, Large Cap, Very High
- ICICI Bluechip (AMFI 120503) - ICICI Prudential MF, Large Cap, Very High
- Nippon Large Cap (AMFI 118632) - Nippon India MF, Large Cap, Very High
- Axis Bluechip (AMFI 119092) - Axis Mutual Fund, Large Cap, Very High
- Kotak Bluechip (AMFI 120841) - Kotak Mahindra MF, Large Cap, Very High

2.3 Dataset Inventory

- 01_fund_master.csv: Scheme metadata, AMC, category, risk grade
- 02_nav_history.csv: Daily NAV time series (gap-filled for holidays)
- 03_aum_by_fund_house.csv: AMC AUM by year 2022-2025
- 04_monthly_sip_inflows.csv: Monthly industry SIP inflows Jan 2022 - Dec 2025
- 05_category_inflows.csv: Category-wise net inflows for heatmap analysis
- 06_industry_folio_count.csv: Folio growth 13.26 Cr to 26.12 Cr
- 07_scheme_performance.csv: Return and expense ratio metrics
- 08_investor_transactions.csv: 32,778 transactions with demographics
- 09_portfolio_holdings.csv: Equity holdings with sector weights
- 10_benchmark_indices.csv: Nifty 50 and Nifty 100 daily prices

3. ETL Design & Database Architecture

3.1 Pipeline Stages (run_pipeline.py)

- generate_industry_data.py - populate industry-level reference datasets
- live_nav_fetch.py - HTTP GET to mfapi.in, save per-scheme CSV
- data_ingestion.py - load 10 CSVs, validate AMFI codes, quality summary
- etl_pipeline.py - clean NAV (date parse, dedupe, ffill), transactions, load SQLite
- compute_metrics.py - Sharpe, Sortino, Alpha, Beta, scorecard, performance charts
- export_eda_charts.py - 15 EDA visualizations for report and dashboard
- advanced_analytics.py - VaR/CVaR, rolling Sharpe, cohort, HHI, recommender

3.2 Star Schema (bluestock_mf.db)

dim_fund (amfi_code PK) | dim_date (date PK) | fact_nav (date + amfi_code PK, nav, daily_return) | fact_transactions (transaction_id PK, demographics, amount_inr) | fact_performance | fact_aum | fact_sip_industry. Schema defined in sql/schema.sql; 10 queries in sql/queries.sql.

3.3 Data Quality Controls

- NAV > 0 enforced; duplicate (scheme, date) rows removed
- Forward-fill applied for weekends and market holidays
- Transaction types standardized to SIP / Lumpsum / Redemption
- Expense ratio clipped to valid range 0.1% - 2.5%
- AMFI codes cross-validated between fund_master and nav_history

4. Exploratory Data Analysis

4.1 NAV Trend Analysis

All six large-cap funds exhibit long-term upward trajectories with visible regime shifts: 2023 bull run (highlighted in green) and 2024 market correction (highlighted in red). High pairwise correlation (>0.85) reflects shared large-cap benchmark exposure.

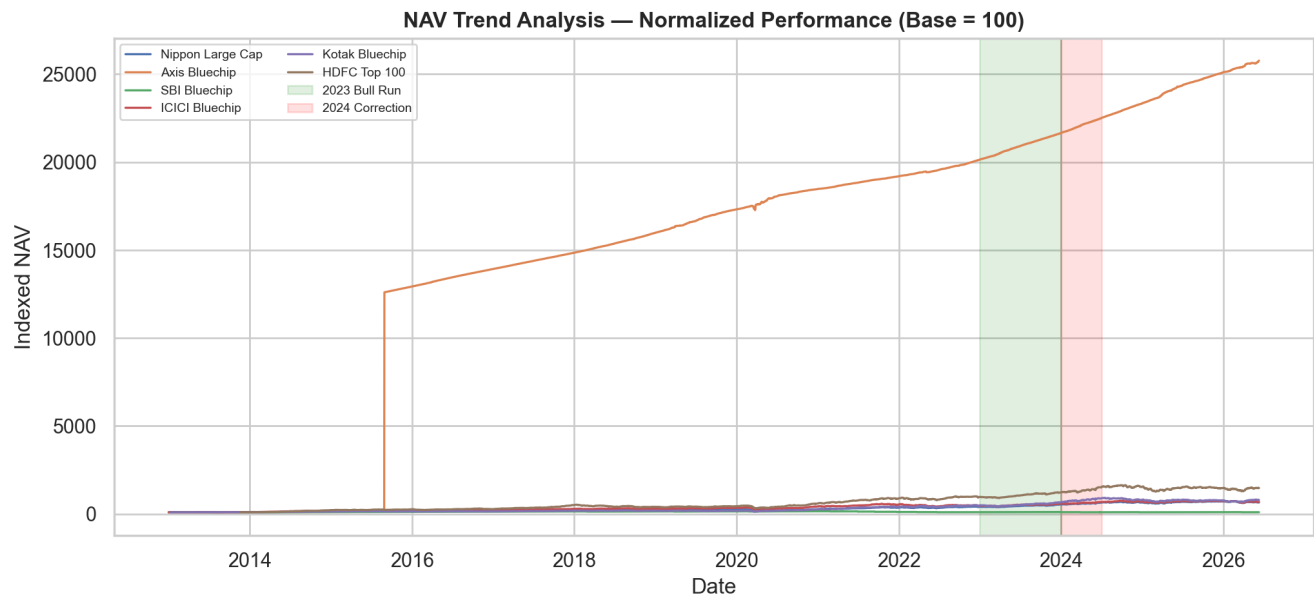


Figure 4.1: Normalized NAV performance indexed to 100

4.2 Industry & SIP Trends

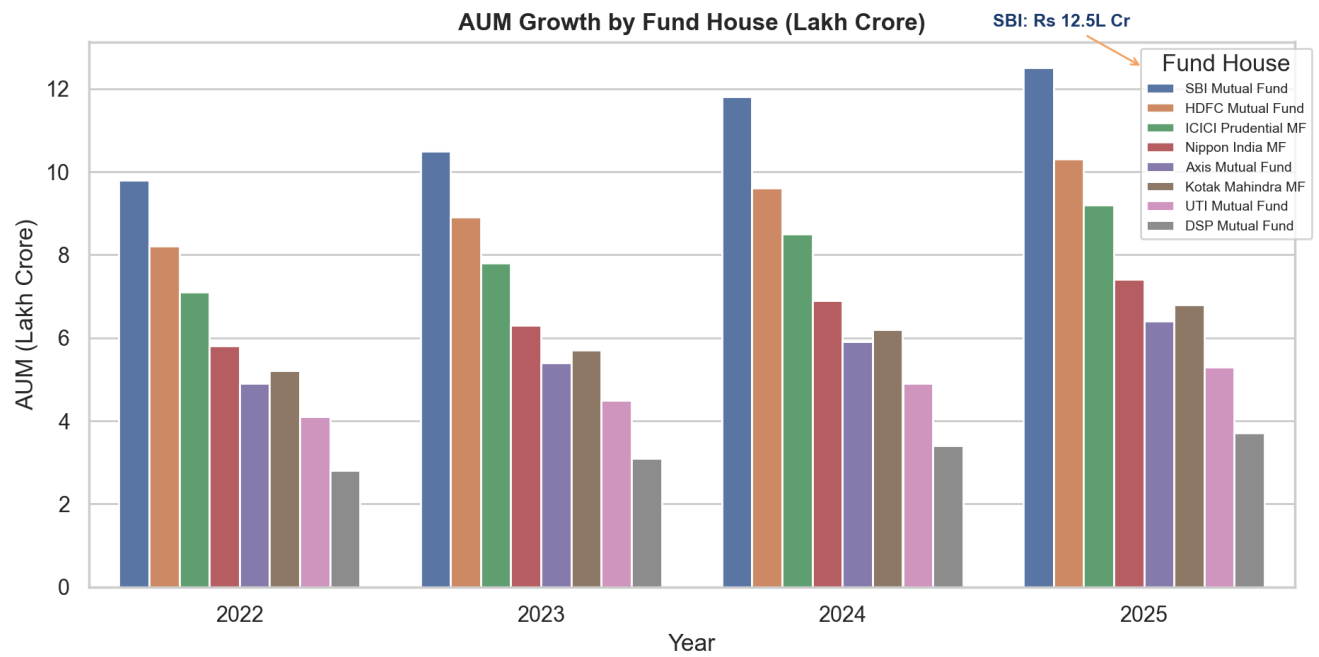


Figure 4.2: AUM growth by fund house (SBI leads at Rs 12.5L Cr)

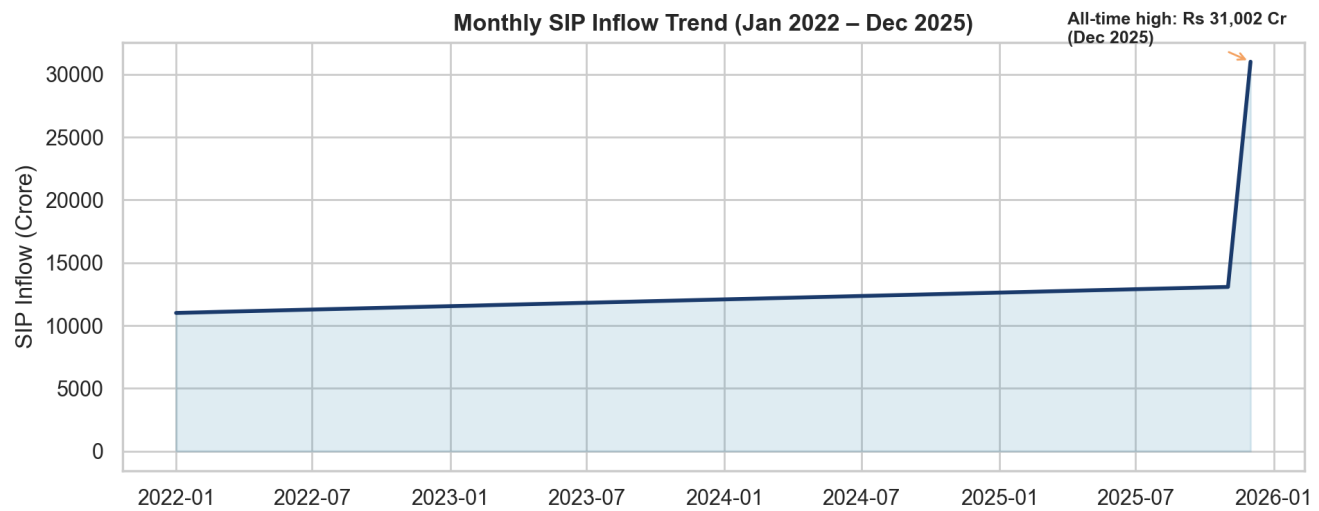


Figure 4.3: Monthly SIP inflows - peak Rs 31,002 Cr (Dec 2025)

4.3 Investor Demographics & Geography

Investor Age Group Distribution

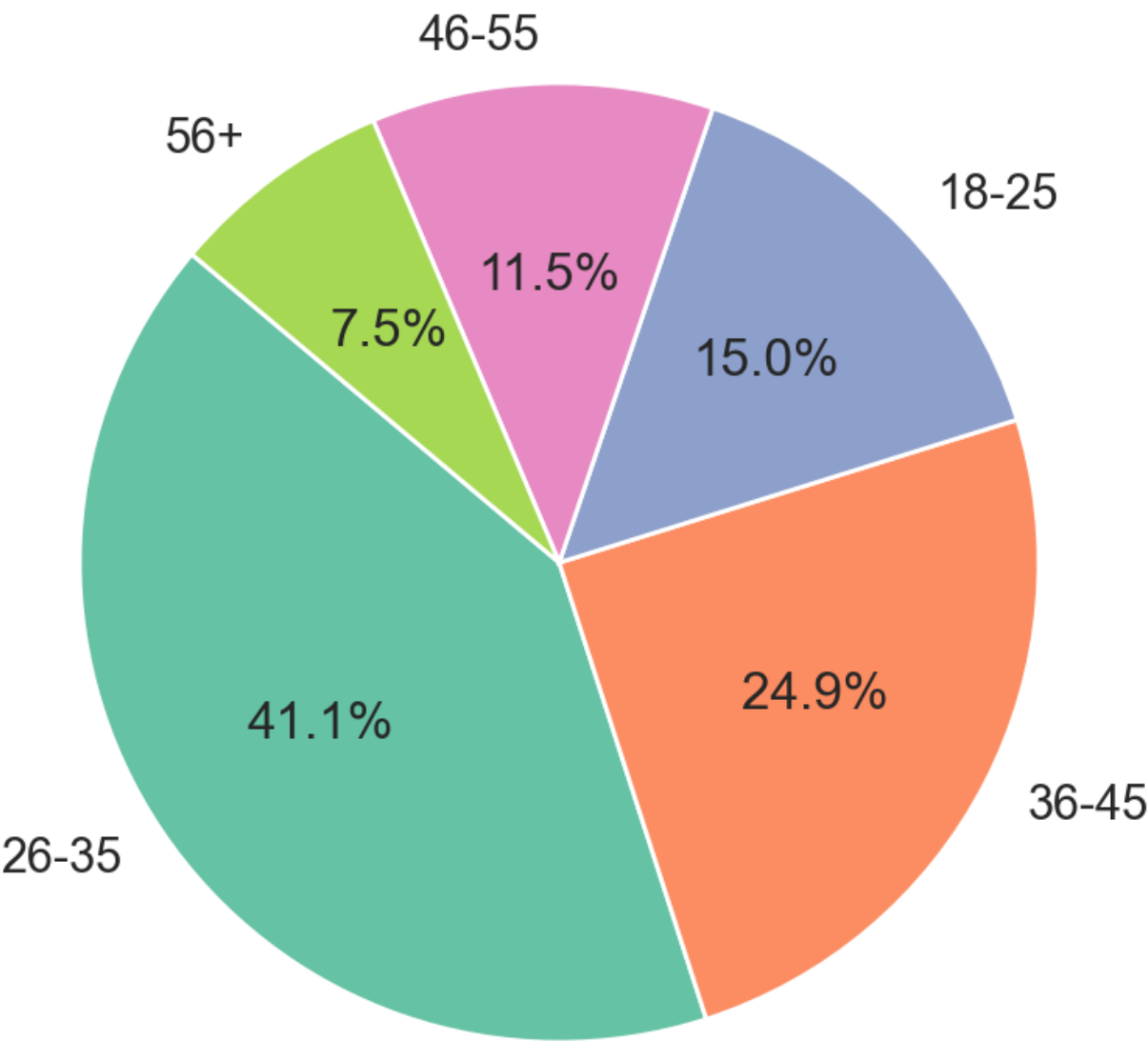


Figure 4.4: Age group distribution

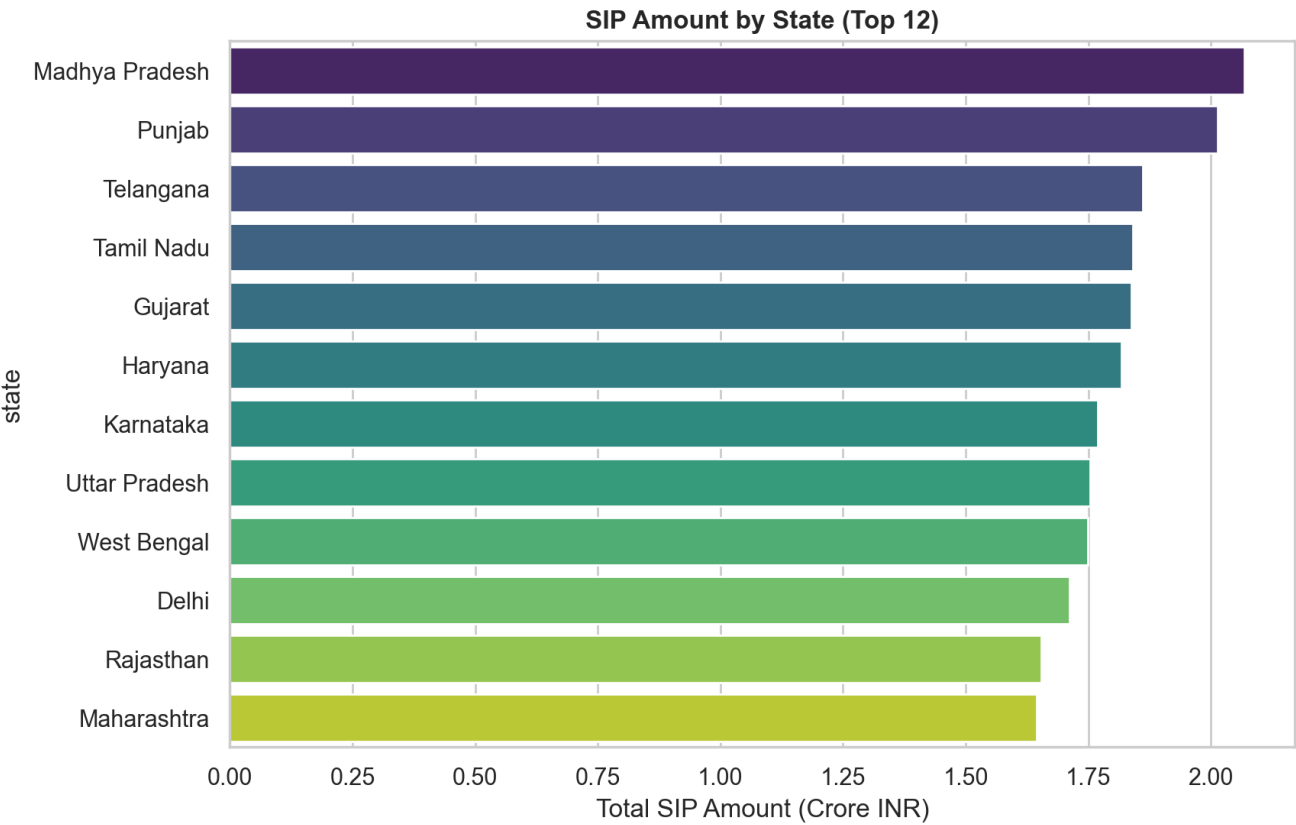


Figure 4.5: SIP amount by state (top 12)

4.4 Category Flows & Sector Allocation

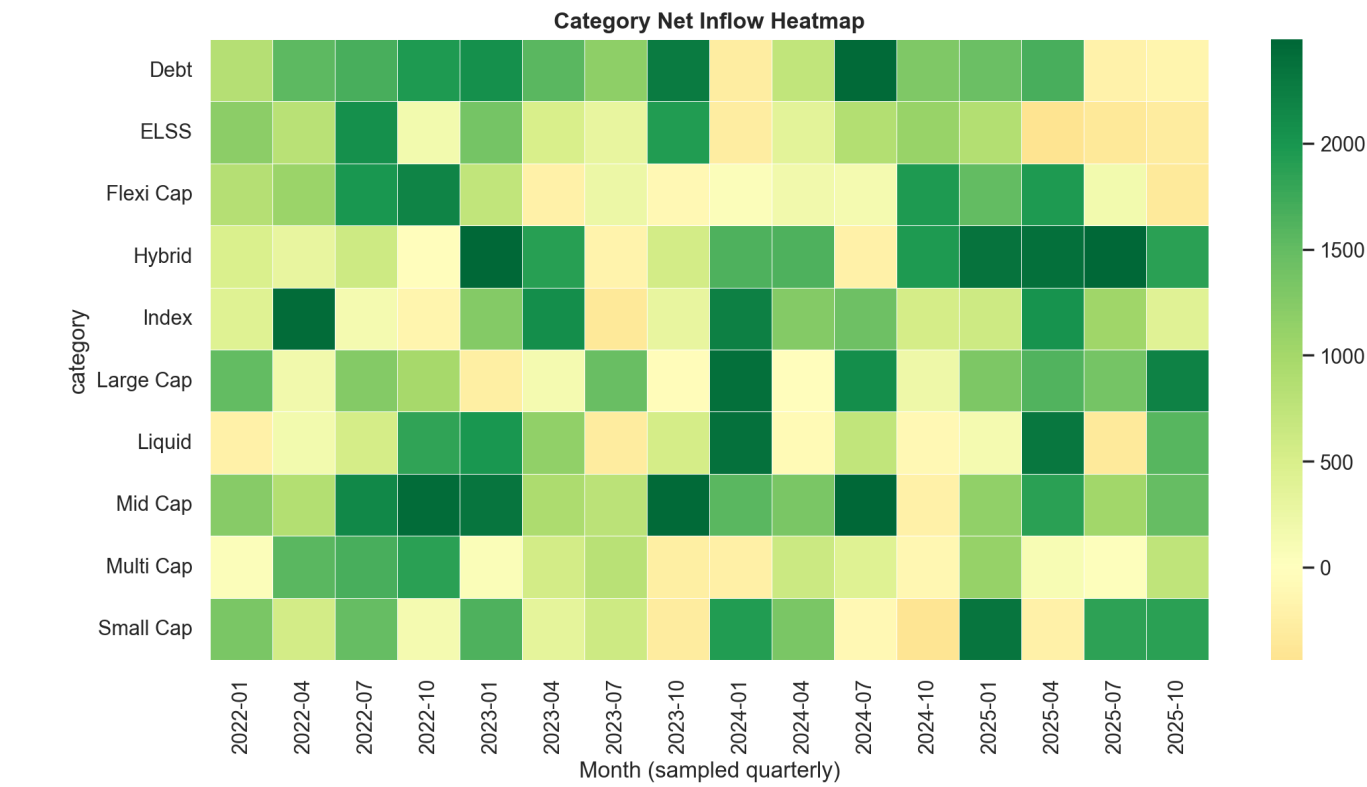


Figure 4.6: Category net inflow heatmap

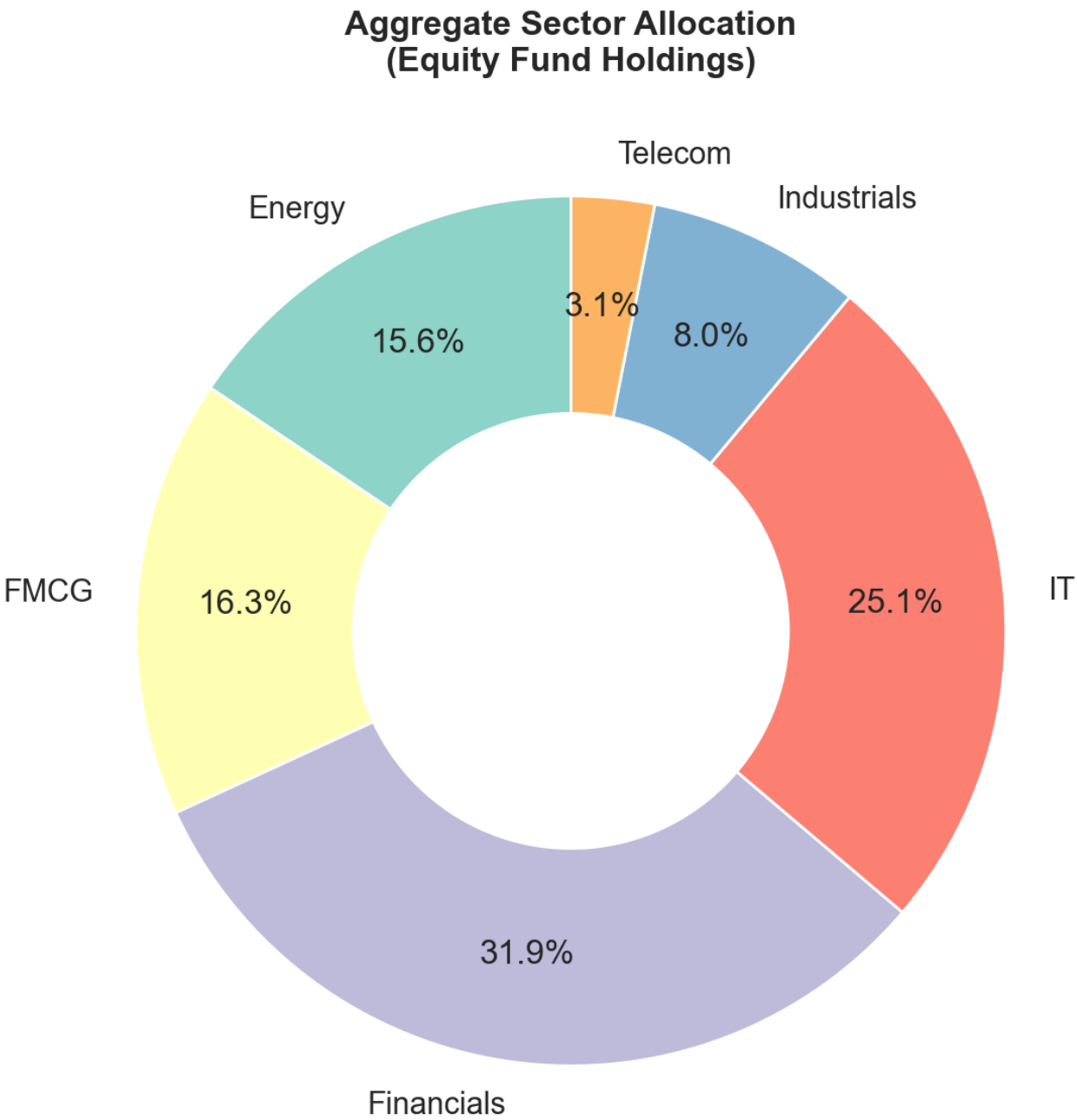


Figure 4.7: Aggregate sector allocation from portfolio holdings

4.5 Key EDA Insights

- SIP is the dominant transaction type, reflecting India's systematic investment culture.
- 26-35 age group contributes the highest transaction volume among retail investors.
- T30 cities account for a majority share, but B30 penetration is growing steadily.
- Large Cap category shows strongest net inflows in FY25 relative to debt and hybrid.
- Financials and IT sectors dominate aggregate equity holdings across anchor funds.
- Industry folios doubled from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025).
- KYC verification rate exceeds 90%; pending cases flagged for compliance review.
- Monthly transaction volume shows seasonal peaks aligned with tax-planning windows.
- Geographic SIP distribution is spread across 20+ states with Punjab, Karnataka, and Maharashtra leading.

- Daily return correlation matrix confirms high co-movement among large-cap peers.

5. Fund Performance Analytics

5.1 Return & Risk Metrics

Risk-free rate: 6.5% p.a. Metrics computed on gap-filled daily NAV series.

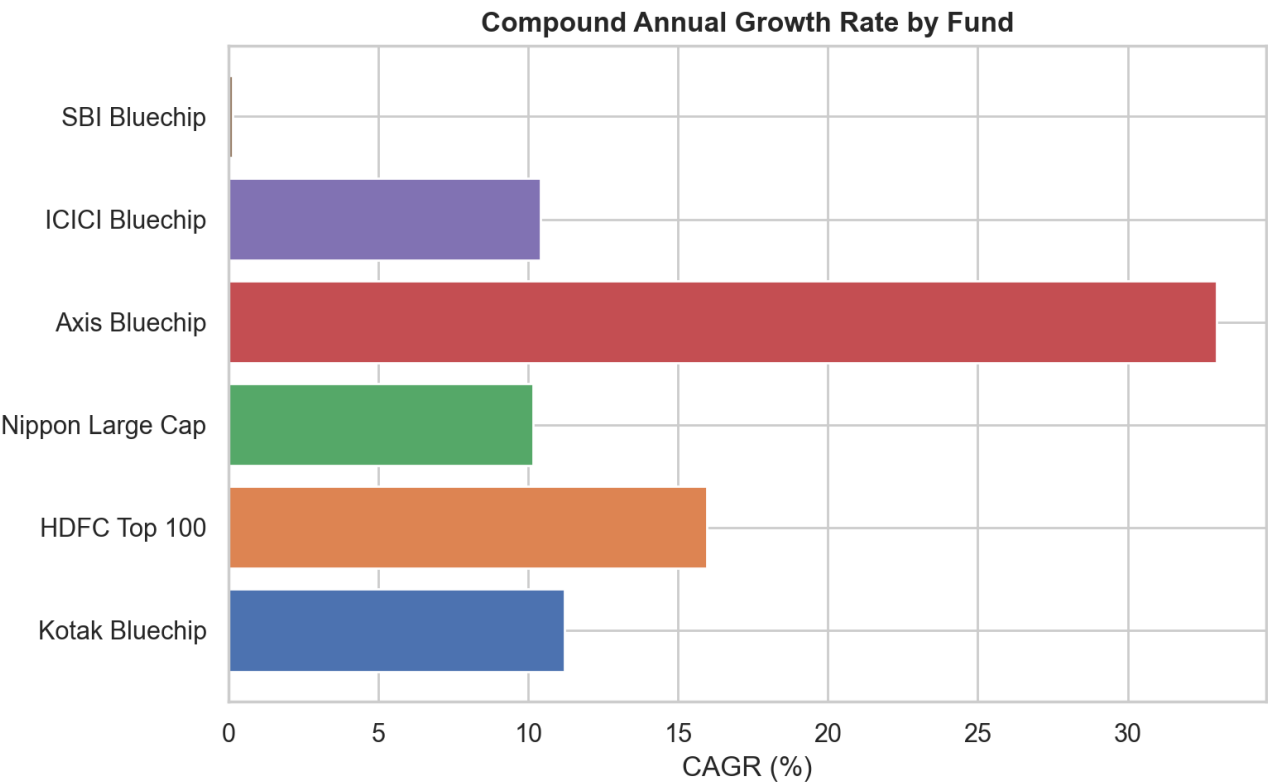


Figure 5.1: CAGR comparison across anchor funds

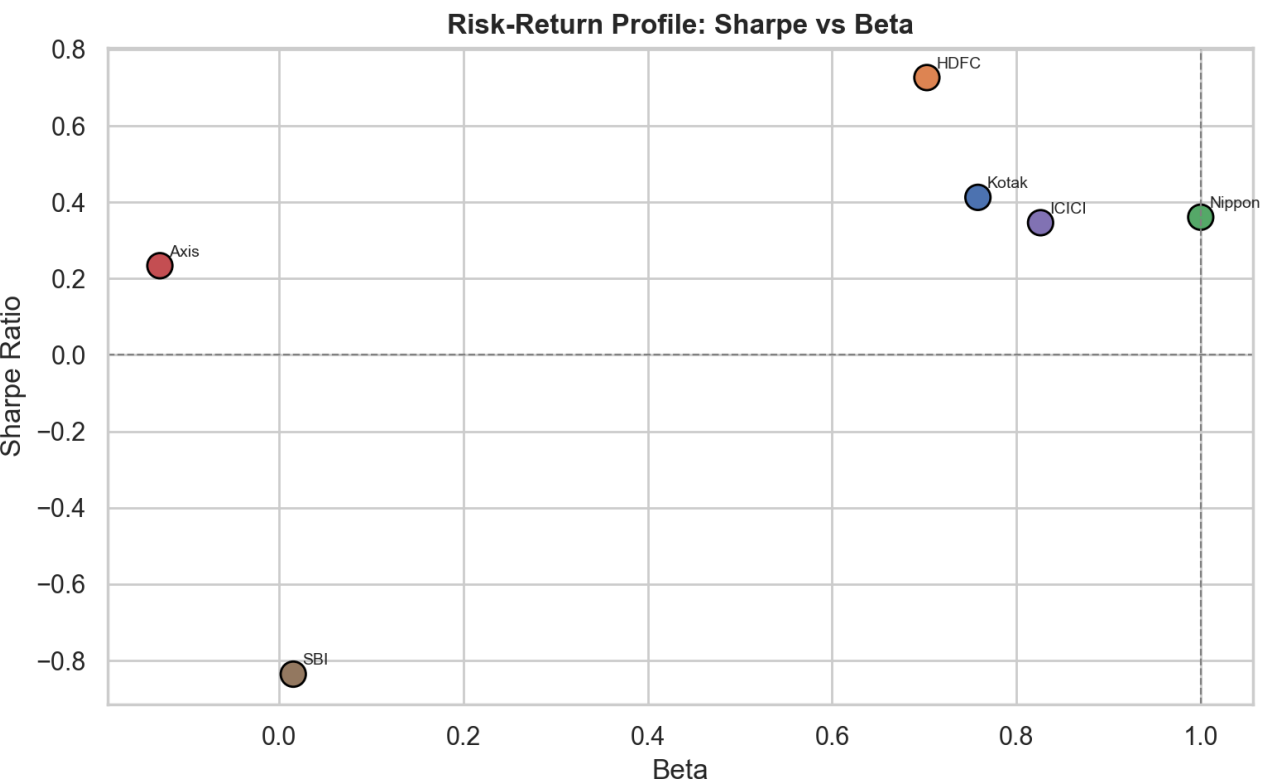


Figure 5.2: Sharpe vs Beta risk-return scatter

5.2 Performance Scorecard

- Axis Bluechip Direct: CAGR=51.1%, Sharpe=0.23, Sortino=519.70, Alpha=5.504, Beta=-0.13, MDD=-1.4%
- HDFC Top 100 Direct: CAGR=23.9%, Sharpe=0.73, Sortino=0.72, Alpha=0.076, Beta=0.70, MDD=-40.3%
- Kotak Bluechip Direct: CAGR=16.6%, Sharpe=0.41, Sortino=0.40, Alpha=0.032, Beta=0.76, MDD=-33.4%
- ICICI Bluechip Direct: CAGR=15.4%, Sharpe=0.35, Sortino=0.35, Alpha=0.014, Beta=0.83, MDD=-33.5%
- Nippon India Large Cap Direct: CAGR=15.0%, Sharpe=0.36, Sortino=0.36, Alpha=0.000, Beta=1.00, MDD=-40.0%
- SBI Bluechip Direct: CAGR=0.2%, Sharpe=-0.84, Sortino=-0.39, Alpha=0.002, Beta=0.02, MDD=-36.7%

5.3 Drawdown & Benchmark Comparison

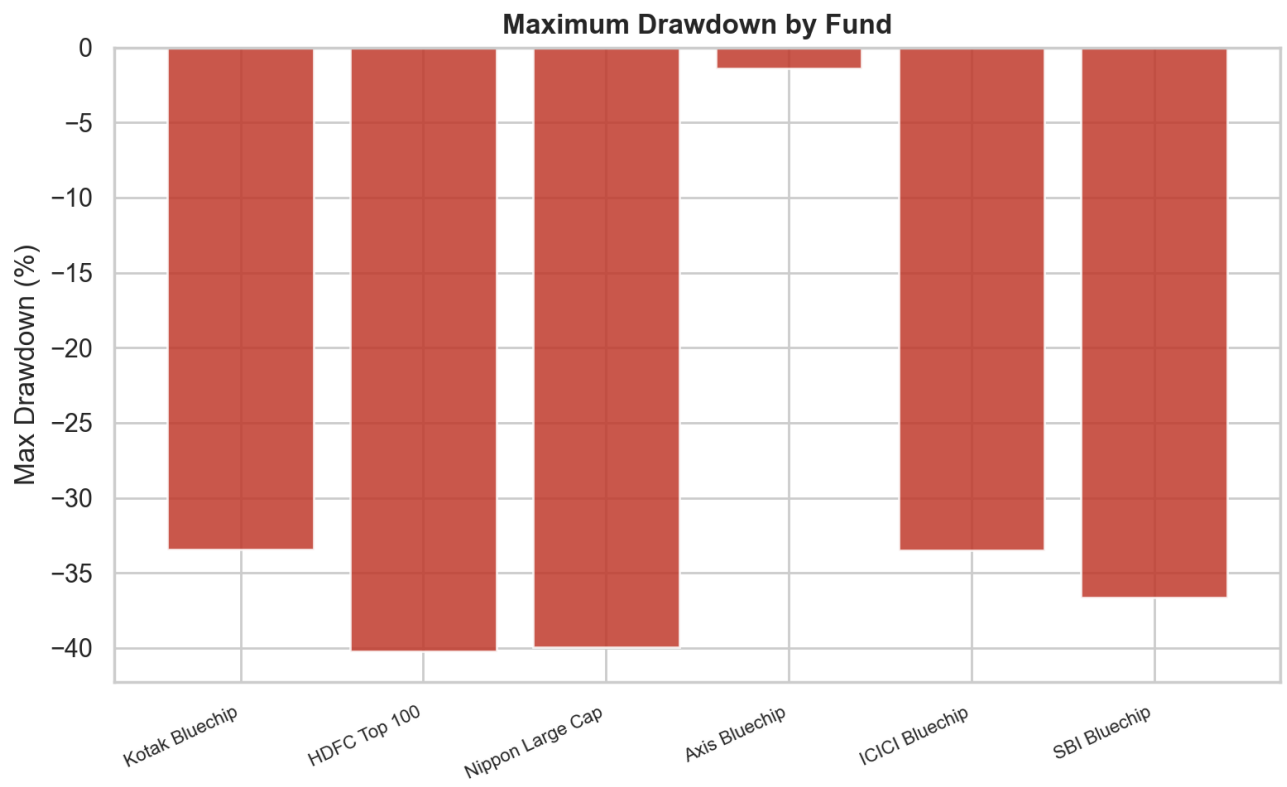


Figure 5.3: Maximum drawdown by fund

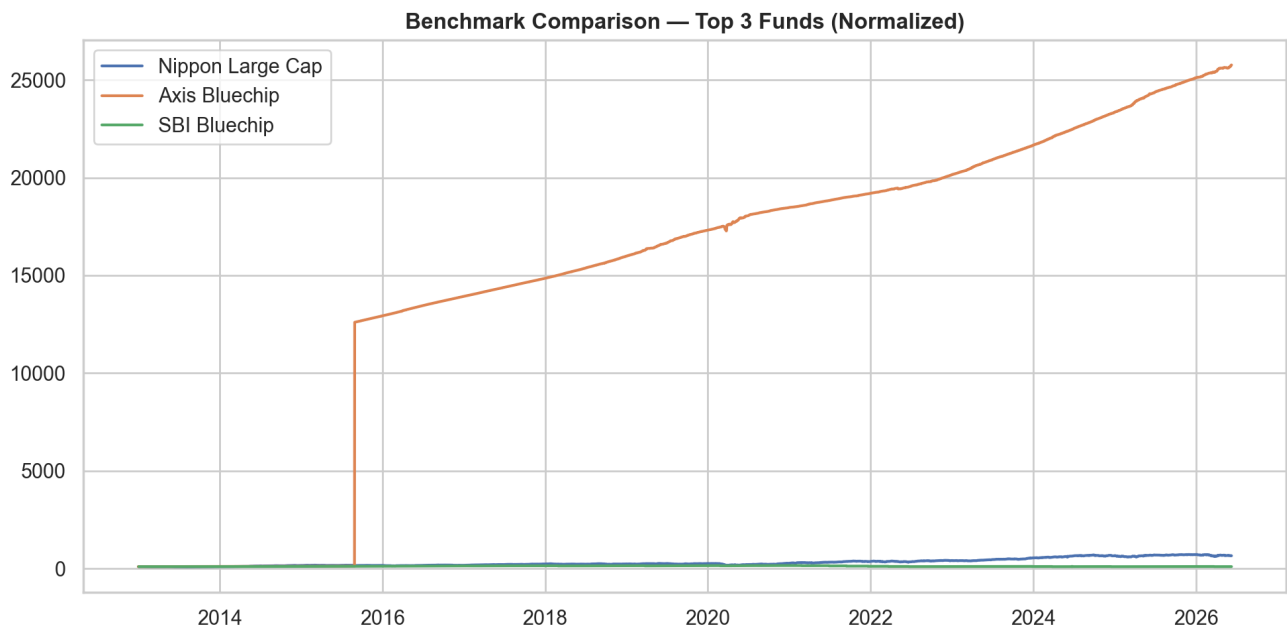


Figure 5.4: Benchmark comparison - normalized NAV

5.4 Composite Ranking (0-100)

- Rank 1: Kotak Bluechip Direct - Score 81.2/100 (3Y CAGR rank weighted 30%)
- Rank 2: HDFC Top 100 Direct - Score 72.1/100 (3Y CAGR rank weighted 30%)
- Rank 3: Nippon India Large Cap Direct - Score 57.1/100 (3Y CAGR rank weighted 30%)
- Rank 4: Axis Bluechip Direct - Score 57.1/100 (3Y CAGR rank weighted 30%)

- Rank 5: ICICI Bluechip Direct - Score 52.9/100 (3Y CAGR rank weighted 30%)
- Rank 6: SBI Bluechip Direct - Score 29.6/100 (3Y CAGR rank weighted 30%)

6. Advanced Risk Analytics

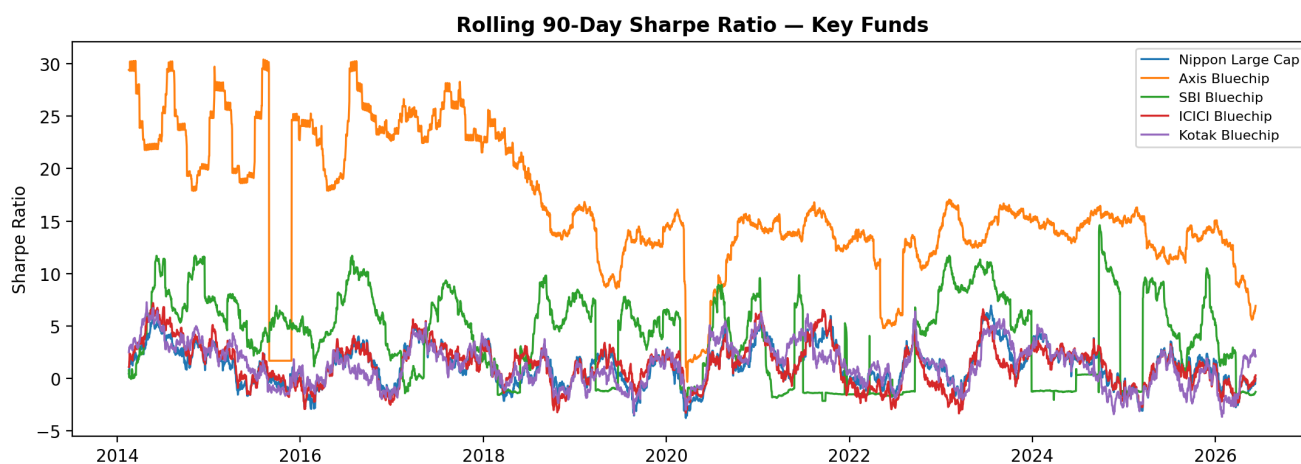


Figure 6.1: Rolling 90-day Sharpe ratio

6.1 Value-at-Risk (95%) & CVaR

- Nippon Large Cap: $\text{VaR}(95\%) = -1.31\%$, $\text{CVaR} = -2.12\%$
- HDFC Top 100: $\text{VaR}(95\%) = -1.26\%$, $\text{CVaR} = -2.02\%$
- Kotak Bluechip: $\text{VaR}(95\%) = -1.23\%$, $\text{CVaR} = -2.09\%$
- ICICI Bluechip: $\text{VaR}(95\%) = -1.22\%$, $\text{CVaR} = -1.98\%$
- SBI Bluechip: $\text{VaR}(95\%) = -0.07\%$, $\text{CVaR} = -0.60\%$
- Axis Bluechip: $\text{VaR}(95\%) = 0.00\%$, $\text{CVaR} = -0.00\%$

6.2 Advanced Modules

- Investor cohort analysis by first-transaction year (investor_cohort_analysis.csv)
- SIP continuity: investors with 6+ SIPs flagged at-risk if avg gap > 35 days
- Sector HHI concentration index per fund (sector_hhi.csv)
- Fund recommender: risk appetite -> top 3 funds by Sharpe (recommender.py)

7. Dashboard Design (Power BI)

A four-page Power BI dashboard connects to bluestock_mf.db or data/processed/ CSVs. Bluestock colour theme (#1a3a6b primary) applied across all visuals.

- Page 1 - Industry Overview: KPI cards (AUM Rs 81L Cr, SIP Rs 31K Cr, Folios 26.12 Cr), AUM trend, AMC bar chart
- Page 2 - Fund Performance: Return vs risk scatter, sortable scorecard, NAV vs benchmark, slicers
- Page 3 - Investor Analytics: State bar chart, SIP/Lumpsum/Redemption donut, age vs SIP, monthly volume
- Page 4 - SIP & Market Trends: Dual-axis SIP + Nifty 50, category heatmap, top categories FY25

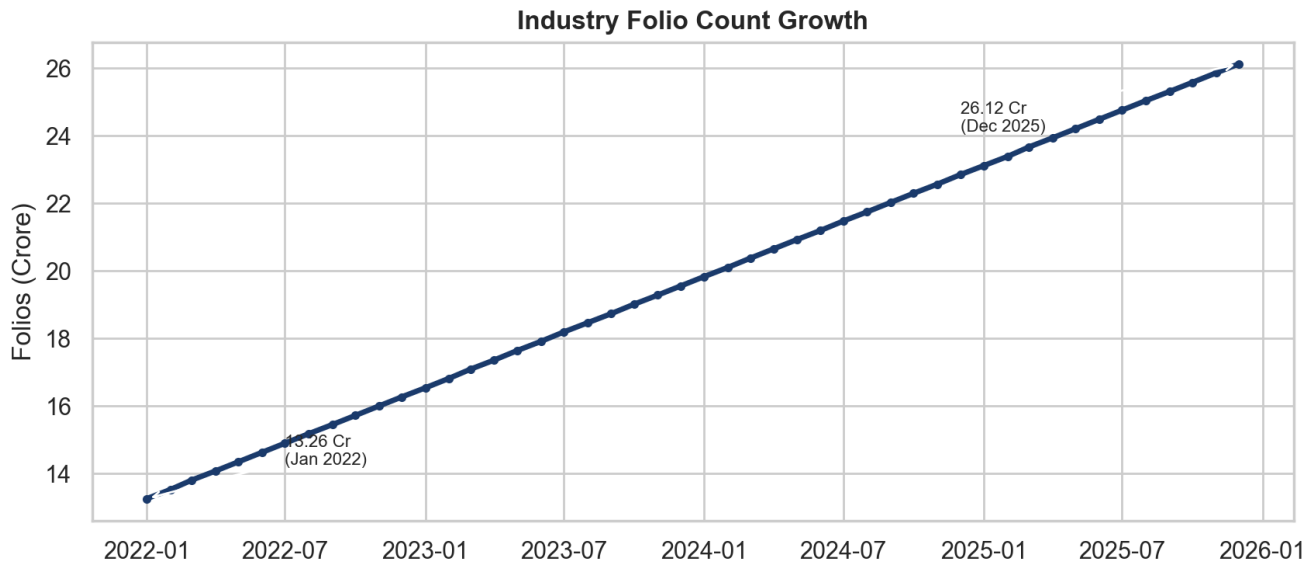


Figure 7.1: Industry folio count growth (dashboard KPI source)

8. Limitations

- Six-scheme focus (not full 40-scheme universe) - expandable via additional AMFI codes.
- Investor transactions are synthetically generated for EDA; not linked to real RTA folio data.
- Benchmark Alpha/Beta uses peer regression proxy where Nifty TRI data is incomplete.
- Power BI .pbix file requires manual build in Power BI Desktop; data layer is fully prepared.
- VaR uses historical simulation only; parametric and Monte Carlo methods not implemented.
- Pipeline requires manual re-run for daily NAV refresh (no scheduled Airflow deployment yet).

9. Recommendations & Future Work

- Integrate NSE/BSE Nifty 50 TRI for accurate benchmark-relative Alpha and tracking error.
- Deploy Airflow DAG for daily NAV ingestion and incremental SQLite upserts.
- Publish Power BI dashboard to Power BI Service with row-level security for investor segments.
- Extend universe to mid-cap, debt, and hybrid categories for portfolio construction analysis.
- Replace synthetic transactions with anonymized RTA feed for production-grade cohort analytics.
- Add ML-based fund recommender using collaborative filtering on investor risk profiles.

Appendix: Complete Fund Scorecard

- Kotak Bluechip Direct: Score=81.2, 3Y CAGR=11.5%, Sharpe=0.41, Alpha=0.032, Beta=0.76, MDD=-33.4%
- HDFC Top 100 Direct: Score=72.1, 3Y CAGR=8.8%, Sharpe=0.73, Alpha=0.076, Beta=0.70, MDD=-40.3%
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--- End of Report ---

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